

## 02-510/710: Computational Genomics, Spring 2026

### HW1: Genomic data processing Differential Equation

*Version: 1*

*Due: 23:59 EST, Feb 3, 2026 on Canvas*

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**Topics** in this assignment:

1. Compression of genomic data
2. Normalization
3. Differential gene expression analysis
4. Probabilistic modeling

**What to hand in.**

- One write-up (in pdf format) addressing each of following questions.
- All source code. If the skeleton is provided, you just need to complete the script and send it back. Your code is tested by autograder, please be careful with your main script name and output format.

Submit the following file which contain the completed code and the pdf file to Canvas separately.

./S2026HW1.pdf  
./normalization.py  
./de\_genes.py  
./sc\_counts.h5ad

**Please note that all the solutions and code must be your own. We will check for plagiarism after the final submission.**

1. [30 points] Quantifying compression of genomic sequences

In this problem, you will manually analyze how much a short DNA sequence can be compressed using different compression schemes discussed in lecture. The goal is to understand the limitations of compression on typical genomic data.

Unless otherwise stated, assume that each nucleotide character (A, C, G, T) is stored using one byte (8 bits) in the uncompressed representation.

**Your task**

- (a) (2 points) Consider the following DNA sequence of length 20:

$$S = \text{ATGATGATGATGCCCCCCC}$$

Compute the total size of this sequence in bytes when stored without compression.

Solution

- (b) (5 points) Apply run-length encoding (RLE) to the sequence  $S$ . Assume that each run is stored as a pair consisting of the character and the run length, and that each character requires 1 byte and each run length requires 1 byte. (Thus each run costs 2 bytes.)

Write down the RLE-encoded form of  $S$  and compute the total compressed size in bytes.

Solution

- (c) (15 points) Now consider applying a greedy Lempel–Ziv (LZ77-style) compression scheme to the same sequence  $S$ . Assume a sliding window size of 20 characters, and that matches of length at least 3 are replaced by a match token; all other characters are encoded as literals.

To simplify counting, use the following size model:

- A literal token stores exactly one character and costs 1 byte.
- A match token stores exactly three fields: **offset**, **length**, and **next**, and each field costs 1 byte (so each match token costs 3 bytes total).

Here, **offset** is the distance backward to the match start in the window, **length** is the number of matched characters, and **next** is the next character after the match.

Using greedy parsing (always choose the longest available match at the current position), identify the match token(s) that will be used, and compute the total compressed size in bytes.

For a general description of the LZ77 compression scheme, see for example:

- Wikipedia: [https://en.wikipedia.org/wiki/LZ77\\_and\\_LZ78](https://en.wikipedia.org/wiki/LZ77_and_LZ78)

Solution

- (d) (8 points) Compare the compressed sizes obtained using RLE and simplified Lempel–Ziv to the uncompressed size. State which method performs better on this sequence and briefly explain why.

Solution

## 2. [40 points] Normalization

In this question, we will implement two different normalization techniques on real RNA-seq data.

### Data Description

We provide data from a study in which a human cell line (A549) was treated with a specific drug (dexamethasone). Samples were collected following treatment to create a time course. For this question you would use 24 samples for which raw RNA-seq data is provided (file: ReadCounts.txt). The table is of size  $n \times m$  where  $n$  is the total number of genes and  $m$  is the total number of samples. We have  $\sim 8K$  genes in the matrix.

You have also been provided with the lengths of each gene present in the data matrix (file: GeneLengths.txt).

### Your task

- (a) (10 points) Normalize the data using two normalization techniques, i) RPKM, ii) size factor normalization.

RPKM normalizes data by the total number of reads and the length of each transcript. This normalization technique, however, is not always effective since few, very highly expressed genes can dominate the total count and skew the expression analysis.

Another normalization technique consists of computing the effective library size by considering a size factor for each sample. By dividing each sample's counts by the corresponding size factors, we bring all the count values to a common scale, making them comparable. Intuitively, if sample A is sequenced  $N$  times deeper than sample B, the read counts of non-differentially expressed genes are expected to be, on average,  $N$  times higher in sample A than in sample B, even if there is no difference in expression.

If we expect that most genes are not different between the samples, we can use the median ratio to estimate these factors. Thus, to estimate the size factors for each sample, we take the median of the ratios of observed counts  $k_{ij}$  to their geometric mean.

$$s_j = \operatorname{median}_i \frac{k_{ij}}{(\prod_{v=1}^m k_{iv})^{\frac{1}{m}}}$$

After computing the size factors, we can normalize the read counts by dividing each sample by their corresponding size factors.

$$\hat{k}_{ij} = \frac{k_{ij}}{s_j}$$

You have been provided with a skeleton python script *normalization.py*. Complete the functions *rpkm*, *size\_factor* in the *normalization.py* script.

**Note:** Your code is graded by an autograder. The script should be able to run with command line:

```
python normalization.py ReadCounts.txt GeneLengths.txt
```

All the normalization functions should return a  $n \times m$  array.

- (b) (10 points) Submit the boxplots of the log2 converted count data for each sample for the i) raw counts, ii) RPKM normalized counts, iii) size factor normalized counts.

Note: Your plots will not be graded by the autograder. Include the plots in the pdf file.

Solution

- (c) (10 points) Comparing the 3 boxplots, which normalization technique would you choose for this dataset? Explain.

Solution

- (d) (10 points) Let  $R(g_1)$  and  $R(g_2)$  be the raw read counts (number of reads mapped to a gene) for genes  $g_1$  and  $g_2$  in a sample. And  $N(g_1)$  and  $N(g_2)$  be the normalized read counts for genes  $g_1$  and  $g_2$  in a sample. Suppose  $R(g_1) > R(g_2)$ , for the following questions choose ALL answers that could be correct and briefly explain your answer.

i. If we use RPKM normalization then:

- A.  $N(g_1) > N(g_2)$
- B.  $N(g_1) = N(g_2)$
- C.  $N(g_1) < N(g_2)$

Solution

ii. If we use scale factor normalization then:

- A.  $N(g_1) > N(g_2)$
- B.  $N(g_1) = N(g_2)$
- C.  $N(g_1) < N(g_2)$

Solution

### 3. [30 points] Differential gene expression analysis

In this exercise we will complete steps to identify differentially expressed genes from the size factor normalized read count data you saved as 'size\_factor\_normalized\_counts.txt', from Q2. We will also learn how to perform multiple hypothesis testing to identify significant differentially expressed genes.

You have been supplied with the numerical sample labels in *labels.txt*. There are two types of samples, 1) cell treated with dexamethasone(case) and 2) cell treated with ethanol (control). The gene names have been provided in *GeneNames.txt* file.

**Note:** Your code for this problem is graded by an autograder. The script should be able to run with command line:

```
python de_genes.py
```

Include requested plots in the pdf file.

#### Your task

- (a) (10 points) Dispersion of gene  $i$  in condition  $C$ ,  $C \in \{Case, Control\}$  can be computed as

$$disp_i^C = \frac{std(\hat{k}_{i,C})}{mean(\hat{k}_{i,C})}$$

Submit a log2-log2 scatterplot of dispersion vs mean in the size factor counts dataset, for the two conditions (but with all the data on one plot). Use different colors for case and control condition. How would you interpret this plot, and why?

Solution

- (b) (10 points) Compute the log2 fold change of the genes in the two conditions. Select top 10 upregulated genes, and top 10 downregulated genes, and submit a gene expression heatmap for the selected genes. Place both upregulated and downregulated genes on the same heatmap. Are these the most useful genes to investigate? Why or why not?

Solution

- (c) (10 points) Determining whether the gene expressions in two conditions are statistically different consists of rejecting the null hypothesis that the two data samples come from distributions with equal means. To do this, we can calculate a p-value for each gene.

However, thresholding P-values to determine what fold changes are more significant than others is not appropriate for this type of data analysis, due to the multiple hypothesis testing problem. When performing a large number of simultaneous tests, the probability of getting a significant result simply due to chance increases with the number of tests. In order to account for multiple testing, perform a correction (or adjustment) of the P-values so that the probability of observing at least one significant result due to chance remains below the desired significance level. We can use The Benjamini-Hochberg (BH) adjustment.

The BH is defined as:

Let  $p_1 \leq \dots \leq p_n$  be ordered p-values. Define

$$k = i : p_i \leq \frac{i}{n} \alpha$$

and reject  $H_0^1 \dots H_0^k$ .  $\alpha$  is the false discovery rate(FDR) to be controlled. If no such  $i$  exists, then no hypothesis will be rejected.

**Implement the BH correction function in `de_genes.py`.** The implementation must be your own.

#### 4. [40 points] Probabilistic modeling of single-cell RNA-seq count data

In this exercise, we will study probabilistic models for single-cell RNA-seq count data and examine the underlying assumptions of these models. We will derive a closed-form maximum likelihood estimator under a Poisson model with library size normalization, and then introduce the Negative Binomial (NB) model as a commonly used extension that allows for overdispersion. Building on these modeling ideas, we will explore how model-motivated normalization methods aim to mitigate the effect of sequencing depth in practice, using simple empirical diagnostics on a real single-cell RNA-seq dataset.

You are provided with a subset of a human intestine single-cell RNA-seq dataset stored in *sc\_counts.h5ad*. You may use either R or Python for the programming part.

##### Reference: Closed-form distributions used in this problem

- **Poisson distribution.** A random variable  $Y \sim \text{Poisson}(\mu)$  has probability mass function

$$p(Y = y) = \frac{\mu^y e^{-\mu}}{y!}, \quad y = 0, 1, 2, \dots$$

with

$$\mathbb{E}[Y] = \mu, \quad \text{Var}(Y) = \mu.$$

- **Gamma distribution.** A random variable  $Z \sim \text{Gamma}(k, \theta)$  with shape parameter  $k > 0$  and scale parameter  $\theta > 0$  has density

$$p(Z = z) = \frac{1}{\Gamma(k)\theta^k} z^{k-1} e^{-z/\theta}, \quad z > 0,$$

with

$$\mathbb{E}[Z] = k\theta, \quad \text{Var}(Z) = k\theta^2.$$

- **Negative Binomial distribution (mean–dispersion form).** A random variable  $Y \sim \text{NB}(\mu, \alpha)$  has probability mass function

$$p(Y = y) = \frac{\Gamma(y + \alpha^{-1})}{\Gamma(\alpha^{-1}) y!} \left( \frac{\alpha^{-1}}{\alpha^{-1} + \mu} \right)^{\alpha^{-1}} \left( \frac{\mu}{\alpha^{-1} + \mu} \right)^y, \quad y = 0, 1, 2, \dots$$

with

$$\mathbb{E}[Y] = \mu, \quad \text{Var}(Y) = \mu + \alpha\mu^2.$$

- **Connection via Poisson–Gamma mixture.** If

$$Y \mid Z \sim \text{Poisson}(Z), \quad Z \sim \text{Gamma}(k, \theta),$$

then the marginal distribution of  $Y$  is Negative Binomial with

$$\mu = k\theta, \quad \alpha = \frac{1}{k}.$$

##### Your task



(a) (15 points) **Poisson model with library size normalization**

Assume a gene-wise Poisson model for the observed counts:

$$Y_{g,c} \sim \text{Poisson}(\mu_{g,c}), \quad \mu_{g,c} = L_c \cdot \lambda_g,$$

where  $Y_{g,c}$  denotes the count of gene  $g$  in cell  $c$ ,  $L_c$  is the library size (total counts) of cell  $c$ , and  $\lambda_g$  is the relative expression rate of gene  $g$ .

You may directly use properties of the Poisson distribution listed in the reference below.

- Write down the gene-wise log-likelihood function for  $\lambda_g$ , ignoring terms that do not depend on  $\lambda_g$ .
- By differentiating the log-likelihood, obtain the maximum likelihood estimator (MLE)  $\hat{\lambda}_g$ .
- Using the reference, state the mean and variance of  $Y_{g,c}$  under this model and describe the implied mean–variance relationship.
- In one or two sentences, explain why this mean–variance relationship may not hold in real single-cell RNA-seq data.

Solution

(b) (10 points) **Negative Binomial model**

Single-cell RNA-seq data are often overdispersed relative to the Poisson model. A commonly used alternative is the Negative Binomial (NB) distribution, which can be written as:

$$Y_{g,c} \sim \text{NB}(\mu_{g,c}, \alpha_g),$$

with

$$\mathbb{E}[Y_{g,c}] = \mu_{g,c}, \quad \text{Var}(Y_{g,c}) = \mu_{g,c} + \alpha_g \mu_{g,c}^2,$$

where  $\alpha_g > 0$  is a gene-specific dispersion parameter. Assume the same mean structure as in Part (a):

$$\mu_{g,c} = L_c \cdot \lambda_g.$$

You may directly use results from the reference below.

- Explain how the NB model differs from the Poisson model in terms of the mean–variance relationship. What qualitative effect does the dispersion parameter  $\alpha_g$  have?
- The NB distribution can be derived as a Poisson–Gamma mixture (see reference). In a few sentences, explain *conceptually* how introducing a Gamma-distributed latent rate leads to overdispersion in the observed counts.

Solution

(c) (15 points) **Normalization and sequencing depth effects**

In Parts (a) and (b), we introduced probabilistic models for single-cell RNA-seq count data and discussed how variability beyond library size can arise. One practical goal of such modeling is to motivate normalization procedures that mitigate the effect of sequencing depth on observed expression values.

In practice, Pearson residual-based normalization methods use this idea to transform raw count data into approximately variance-stabilized values, with the goal of reducing systematic dependence on sequencing depth and other technical effects. Such methods are motivated by

count-based models, but are often used as practical normalization tools rather than as exact likelihood-based fits.

In this part, you will work with a subset of a human intestine single-cell RNA-seq dataset provided as `sc_counts.h5ad`, and compare normalization approaches using either **Scanpy** (Python) or **Seurat** (R).

- If using **Scanpy**, you may apply Pearson residual-based normalization using `scanpy.experimental.pp.normalize_pearson_residuals`.
- If using **Seurat**, you may apply `sctransform`, which computes Pearson residual-like normalized values using a regularized count model.

In addition, you should compute standard log-normalized expression values for comparison.

- i. Load the dataset `sc_counts.h5ad` and compute the library size  $L_c$  for each cell.
- ii. Apply standard log-normalization to the count matrix.
- iii. Apply either Pearson residual-based normalization (Scanpy) or `sctransform` (Seurat).
- iv. For each normalization method, compute the *gene-wise* correlation between normalized expression values and library size  $L_c$ , i.e., for each gene  $g$ , compute the correlation across cells between the vector of normalized expression values  $\{X_{g,c}\}_c$  and  $\{L_c\}_c$ .
- v. Summarize the distribution of gene-wise correlations (e.g., using a histogram or boxplot), and briefly compare the results for the two normalization methods.
- vi. Comment on how these observations relate to the goal of mitigating sequencing depth effects.

Solution